

Supplementary Material

Structured Inter-Agent Handoffs and Validation: A Four-Arm Mechanism Study in Multi-Agent Code Generation

This supplement assembles every table and figure cited from the main manuscript. Cells and regimes are not pooled; captions state the relevant denominator, scope, and interpretive limits.

Arm names. The main text uses reader labels; some supplementary tables and figures print the code identifiers. baseline = Single-call; naive = NL handoff; structured_no_validation (or “structured”) = JSON handoff; contract = Contract. The contrast “structure” is JSON handoff minus NL handoff, “enforcement” is Contract minus JSON handoff, and “total” is Contract minus NL handoff.

Table S1. Complete paired contrasts for the six model-study cells

All 24 paired contrasts: 18 mechanism rows (six model-study cells by the three mechanism contrasts) and 6 descriptive system-reference rows, kept in a separate block. Each row gives the task-level difference in percentage points with its 95% percentile interval from 10000 task-cluster bootstrap resamples, and the number of tasks favouring each side. Direction follows the source definition `arm_a` minus `arm_b`, printed on every row: a positive difference favours `arm_a` and a negative one favours `arm_b`. Cells are never pooled across models or task regimes, and the intervals share bootstrap resamples within a cell, so they are not independent across contrasts and are not combined. The system-reference rows compare the multi-agent naive flow with the single-agent baseline, which issues one model call per task against naive's two or more; the two arms do not share a compute budget, so that block is descriptive only and is not an equal-compute control and not a causal mechanism effect.

Block	Model-study cell	Contrast	Direction (<code>arm_a</code> – <code>arm_b</code>)	Difference (pp)	95% CI (pp)	Favourable / equal / adverse
Mechanism contrasts	Study 1A · Gemini	Structure	<code>structured_no_validation</code> – naive	+2.7	[-0.7, +6.7]	4 / 45 / 1
Mechanism contrasts	Study 1A · DeepSeek	Structure	<code>structured_no_validation</code> – naive	+3.3	[-1.3, +8.0]	7 / 41 / 2
Mechanism contrasts	Study 1B · Gemini	Structure	<code>structured_no_validation</code> – naive	+0.7	[-2.7, +4.7]	2 / 46 / 2
Mechanism contrasts	Study 1B · Luna	Structure	<code>structured_no_validation</code> – naive	+2.7	[-1.3, +8.0]	4 / 44 / 2
Mechanism contrasts	Study 2 · Gemini	Structure	<code>structured_no_validation</code> – naive	+3.3	[-1.3, +7.3]	7 / 42 / 1
Mechanism contrasts	Study 2 · Luna	Structure	<code>structured_no_validation</code> – naive	+0.7	[-6.7, +7.3]	8 / 36 / 6
Mechanism contrasts	Study 1A · Gemini	Enforcement	<code>contract</code> – <code>structured_no_validation</code>	-1.3	[-4.7, +1.3]	1 / 47 / 2
Mechanism contrasts	Study 1A · DeepSeek	Enforcement	<code>contract</code> – <code>structured_no_validation</code>	-0.7	[-4.0, +2.7]	3 / 43 / 4
Mechanism contrasts	Study 1B · Gemini	Enforcement	<code>contract</code> – <code>structured_no_validation</code>	+0.7	[-1.3, +2.7]	2 / 47 / 1
Mechanism contrasts	Study 1B · Luna	Enforcement	<code>contract</code> – <code>structured_no_validation</code>	+1.3	[-2.0, +4.7]	4 / 44 / 2
Mechanism contrasts	Study 2 · Gemini	Enforcement	<code>contract</code> – <code>structured_no_validation</code>	0.0	[-5.3, +5.3]	4 / 40 / 6
Mechanism contrasts	Study 2 · Luna	Enforcement	<code>contract</code> – <code>structured_no_validation</code>	+0.7	[-5.3, +6.7]	6 / 38 / 6
Mechanism contrasts	Study 1A · Gemini	Total package	<code>contract</code> – naive	+1.3	[-2.7, +5.3]	3 / 46 / 1
Mechanism contrasts	Study 1A · DeepSeek	Total package	<code>contract</code> – naive	+2.7	[-2.7, +8.0]	8 / 38 / 4
Mechanism contrasts	Study 1B · Gemini	Total package	<code>contract</code> – naive	+1.3	[-2.0, +5.3]	3 / 45 / 2
Mechanism contrasts	Study 1B · Luna	Total package	<code>contract</code> – naive	+4.0	[0.0, +8.7]	7 / 41 / 2
Mechanism contrasts	Study 2 · Gemini	Total package	<code>contract</code> – naive	+3.3	[-2.7, +9.3]	8 / 38 / 4
Mechanism contrasts	Study 2 · Luna	Total package	<code>contract</code> – naive	+1.3	[-6.0, +8.7]	8 / 37 / 5
Descriptive system reference (not an equal-compute control)	Study 1A · Gemini	System reference	naive – baseline	+1.3	[-4.0, +7.3]	2 / 46 / 2
Descriptive system reference (not an equal-compute control)	Study 1A · DeepSeek	System reference	naive – baseline	-6.7	[-14.0, 0.0]	2 / 40 / 8
Descriptive system reference (not an equal-compute control)	Study 1B · Gemini	System reference	naive – baseline	+0.7	[-6.7, +8.0]	4 / 43 / 3
Descriptive system reference (not an equal-compute control)	Study 1B · Luna	System reference	naive – baseline	-8.0	[-15.3, -2.0]	1 / 41 / 8
Descriptive system reference (not an equal-compute control)	Study 2 · Gemini	System reference	naive – baseline	-4.7	[-11.3, +2.0]	4 / 37 / 9
Descriptive system reference (not an equal-compute control)	Study 2 · Luna	System reference	naive – baseline	-8.7	[-17.3, -0.7]	3 / 37 / 10

*Notes. Mechanism contrasts, in the order shown: `contract` – naive, `contract` – `structured_no_validation` and `structured_no_validation` – naive. Each is read as `arm_a` minus `arm_b`. The favourable / equal / adverse triple partitions the 50 tasks of a cell by the sign of the per-task difference. It accompanies the interval as a descriptive companion and is not a separate test. Estimates and interval bounds are rounded to one decimal for display; the full-precision source values are carried in this table's metadata. The system-reference block records `descriptive_only=true`, `equal_compute_control=false` and `causal_interpretation=false` in the source. In plain terms: the difference is real as an observation, but because baseline and naive do not issue the same number of model calls, it cannot be attributed to the coordination mechanism rather than to the added call budget. No *p*-value, significance mark, or ordering appears in this table, and no row pools cells or contrasts.*

Table S2. Exact task-outcome pattern counts

Exact per-cell counts for the five task-outcome patterns, the numerical companion to the main pattern figure: same six model-study cells, same five categories, same source. Each row sums to the 50 tasks of its cell. The categories are mutually exclusive and are assigned from the four arms' pass counts on a task, so a task falls in exactly one of them. This is a post-hoc descriptive summary of observed outcomes; the differences between rows are not a test, and the difference between the Study 1 and Study 2 rows is not a complexity-moderation analysis, since the two studies use different task samples.

Model-study cell	All arms 0/3	All arms 3/3	Same intermediate	Discordant, boundary only	Discordant, with intermediate	Total
Study 1A · Gemini	11	33	0	0	6	50
Study 1A · DeepSeek	7	30	0	0	13	50
Study 1B · Gemini	8	31	1	2	8	50
Study 1B · Luna	8	29	0	1	12	50
Study 2 · Gemini	24	8	0	1	17	50
Study 2 · Luna	21	6	1	0	22	50

Notes. Category definitions: All arms 0/3 — every arm failed all three repetitions; All arms 3/3 — every arm passed all three repetitions; Same intermediate — every arm returned the same intermediate count, either 1/3 or 2/3; Discordant, boundary only — arms disagree but every arm sits at a stable boundary, 0/3 or 3/3; Discordant, with intermediate — arms disagree and at least one arm returned 1/3 or 2/3. Counts are re-derived from the 300 task rows of the source table rather than read from a stored summary, and each row's five categories are checked to sum to 50 before the table is written. Cells are never pooled across models or task regimes and no total or average row is produced; the Total column is the within-row sum only.

Table S3 Panel A. Validator, retry, parse, and compliance detail

Per-cell protocol exposure for the contract arm and the matched structured arm. Each percentage retains its own arm-run denominator. Recovery is conditional on validator activation and is shown as not applicable when no activation occurred. A maximum of three planner attempts means the initial attempt plus at most two bounded retries. These are protocol/process measurements, not task-success outcomes; provider-layer transport retries are separate and absent from this table. Cells are not pooled.

Model-study cell	Contract runs	Structured parse compliance	Contract first-attempt compliance	Validator activations	Recovered after activation	Validation exhausted	Final schema compliance	Mean planner attempts
Study 1A · Gemini	150	150/150 (100.0%)	150/150 (100.0%)	0/150 (0.0%)	N/A (no activation)	0/150 (0.0%)	150/150 (100.0%)	1.000
Study 1A · DeepSeek	150	139/150 (92.7%)	126/150 (84.0%)	24/150 (16.0%)	24/24 (100.0%)	0/150 (0.0%)	150/150 (100.0%)	1.167
Study 1B · Gemini	150	150/150 (100.0%)	150/150 (100.0%)	0/150 (0.0%)	N/A (no activation)	0/150 (0.0%)	150/150 (100.0%)	1.000
Study 1B · Luna	150	150/150 (100.0%)	150/150 (100.0%)	0/150 (0.0%)	N/A (no activation)	0/150 (0.0%)	150/150 (100.0%)	1.000
Study 2 · Gemini	150	150/150 (100.0%)	149/150 (99.3%)	1/150 (0.7%)	1/1 (100.0%)	0/150 (0.0%)	150/150 (100.0%)	1.007
Study 2 · Luna	150	150/150 (100.0%)	150/150 (100.0%)	0/150 (0.0%)	N/A (no activation)	0/150 (0.0%)	150/150 (100.0%)	1.000

Notes. The 25 observed activations are concentrated in Study 1A - DeepSeek (24) and Study 2 - Gemini (1); no pooled table row is created. Structured parse compliance concerns structured_no_validation outputs; first-attempt and final compliance concern contract executions. Recovery after activation cannot be interpreted as recovery of task correctness.

Table S3 Panel B. Validator activation strata and retry-triggering tasks

Panel B of Table S3. Panel A reports how often the validator fired; Panel B reports what the affected runs and tasks looked like afterwards. The upper block splits the contract arm of each Study 1A cell into runs whose plan was rejected at least once (retried) and runs accepted on the first attempt; the lower block reports the same retry-triggering tasks under all four arms. Study 1A - Gemini produced no retried run, so its retried stratum and its retry-triggering task set are reported as NA with the recorded reason no_retried_runs, and no zero-valued estimate is printed for them; its first-attempt row therefore equals the whole contract arm and carries no within-cell contrast. Only the Study 1A - DeepSeek cell carries stratified information. This display is exploratory: strata were not randomized, runs entered the retried stratum because the validator rejected a plan, and no causal validator effect is estimated here or elsewhere. Recovering schema compliance after a rejected plan is a protocol outcome and is not an improvement in task success. Provider-layer transport retries are a separate mechanism and are absent from this panel. Cells are never pooled.

Model-study cell	Stratification block	Stratum or arm	Arm-runs	Base success	Plus success	Retry-triggering tasks	Stratum status
Study 1A · Gemini	Contract arm, retry strata	Retried	NA (no_retried_runs)	NA (no_retried_runs)	NA (no_retried_runs)	0	Not applicable (no_retried_runs)
Study 1A · Gemini	Contract arm, retry strata	First attempt	150	137/150 (91.3%)	114/150 (76.0%)	0	No contrast (equals the whole contract arm)
Study 1A · Gemini	Retry-triggering tasks, all four arms	Baseline	NA (no_retried_runs)	NA (no_retried_runs)	NA (no_retried_runs)	0	Not applicable (no_retried_runs)
Study 1A · Gemini	Retry-triggering tasks, all four arms	Naive	NA (no_retried_runs)	NA (no_retried_runs)	NA (no_retried_runs)	0	Not applicable (no_retried_runs)
Study 1A · Gemini	Retry-triggering tasks, all four arms	Structured (no validation)	NA (no_retried_runs)	NA (no_retried_runs)	NA (no_retried_runs)	0	Not applicable (no_retried_runs)
Study 1A · Gemini	Retry-triggering tasks, all four arms	Contract	NA (no_retried_runs)	NA (no_retried_runs)	NA (no_retried_runs)	0	Not applicable (no_retried_runs)
Study 1A · DeepSeek	Contract arm, retry strata	Retried	24	20/24 (83.3%)	18/24 (75.0%)	18	Stratified
Study 1A · DeepSeek	Contract arm, retry strata	First attempt	126	118/126 (93.7%)	93/126 (73.8%)	18	Stratified
Study 1A · DeepSeek	Retry-triggering tasks, all four arms	Baseline	54	45/54 (83.3%)	42/54 (77.8%)	18	Stratified
Study 1A · DeepSeek	Retry-triggering tasks, all four arms	Naive	54	44/54 (81.5%)	39/54 (72.2%)	18	Stratified
Study 1A · DeepSeek	Retry-triggering tasks, all four arms	Structured (no validation)	54	45/54 (83.3%)	41/54 (75.9%)	18	Stratified
Study 1A · DeepSeek	Retry-triggering tasks, all four arms	Contract	54	45/54 (83.3%)	40/54 (74.1%)	18	Stratified

Notes. The two contract strata partition the 150 contract arm-runs of each cell; DeepSeek splits 24 retried and 126 first-attempt, Gemini 0 and 150. The lower block covers the 18 DeepSeek tasks that triggered at least one validator activation, observed 18 tasks times 3 repeats = 54 arm-runs in each of the four arms; the same tasks are not identified for Gemini because none exists. NA is the only null token in this panel. An NA cell means the quantity is undefined for that stratum, never that it was measured and found to be zero. Percentages are recomputed from the printed counts and are accepted only when they agree with the full-precision rate recorded in the source row.

Table S4. Calls, tokens, latency, and logged cost by arm-run

Descriptive resource and timing detail for all 24 model-study-cell by arm combinations. Included runs are completed result-matched arm-runs; run_error records and unmatched infrastructure calls are excluded from these per-arm summaries and accounted for separately in the source package. Latency median and P95 use R-7 linear interpolation over arm-run latency; observed maximum is the largest recorded arm-run value. Logged cost is provider-reported run-time accounting, not an invoice. Raw tokens, latency, and cost are not compared across models; ratios are calculated only within a cell against naive. Baseline is a descriptive system reference and not an equal-compute control.

Model-study cell	Arm	Included/completed runs (missing usage)	Successful calls/run	Mean total tokens/run	Token ratio to naive	Median latency (s)	P95 latency (s)	Observed max latency (s)	Logged cost/run (USD)	Logged-cost ratio to naive	Pricing snapshot
Study 1A · Gemini	baseline	150/150 (0 missing)	1.000	240.5	0.19×	2.73	3.04	5.11	\$0.000304	0.19×	not recorded
Study 1A · Gemini	naive	150/150 (0 missing)	2.000	1267.2	1.00×	5.91	6.33	9.23	\$0.001597	1.00×	not recorded
Study 1A · Gemini	structured_no_validation	150/150 (0 missing)	2.000	1197.2	0.94×	5.84	6.36	7.38	\$0.001295	0.81×	not recorded
Study 1A · Gemini	contract	150/150 (0 missing)	2.000	1194.0	0.94×	5.80	6.24	7.61	\$0.001281	0.80×	not recorded
Study 1A · DeepSeek	baseline	150/150 (0 missing)	1.000	765.2	0.37×	7.10	60.55	526.17	\$0.000166	0.41×	not recorded
Study 1A · DeepSeek	naive	150/150 (0 missing)	2.000	2091.8	1.00×	22.12	110.82	1218.64	\$0.000409	1.00×	not recorded
Study 1A · DeepSeek	structured_no_validation	150/150 (0 missing)	2.000	2432.6	1.16×	30.33	127.08	408.06	\$0.000479	1.17×	not recorded
Study 1A · DeepSeek	contract	150/150 (0 missing)	2.167	2677.7	1.28×	31.16	221.23	2990.33	\$0.000503	1.23×	not recorded
Study 1B · Gemini	baseline	150/150 (0 missing)	1.000	242.9	0.19×	2.68	3.24	12.11	\$0.000310	0.19×	2026-08-12T11:09:11.442026+00:00
Study 1B · Gemini	naive	150/150 (0 missing)	2.000	1282.9	1.00×	5.78	6.42	8.89	\$0.001622	1.00×	2026-08-12T11:09:11.442026+00:00
Study 1B · Gemini	structured_no_validation	150/150 (0 missing)	2.000	1208.4	0.94×	5.72	6.34	7.53	\$0.001307	0.81×	2026-08-12T11:09:11.442026+00:00
Study 1B · Gemini	contract	150/150 (0 missing)	2.000	1213.7	0.95×	5.69	6.15	7.80	\$0.001315	0.81×	2026-08-12T11:09:11.442026+00:00
Study 1B · Luna	baseline	150/150 (0 missing)	1.000	393.9	0.45×	4.08	10.73	40.89	\$0.000342	0.56×	2026-08-27T08:29:16.936687+00:00
Study 1B · Luna	naive	150/150 (0 missing)	2.000	884.2	1.00×	7.55	13.90	17.56	\$0.000611	1.00×	2026-08-27T08:29:16.936687+00:00
Study 1B · Luna	structured_no_validation	150/150 (0 missing)	2.000	1349.0	1.53×	8.30	15.48	29.58	\$0.000837	1.37×	2026-08-27T08:29:16.936687+00:00
Study 1B · Luna	contract	150/150 (0 missing)	2.000	1369.6	1.55×	8.12	16.28	34.47	\$0.000862	1.41×	2026-08-27T08:29:16.936687+00:00
Study 2 · Gemini	baseline	150/150 (0 missing)	1.000	717.9	0.32×	1.91	2.70	8.34	\$0.001187	0.39×	2026-08-28T14:43:55.703507+00:00
Study 2 · Gemini	naive	150/150 (0 missing)	2.000	2241.7	1.00×	4.90	6.04	7.36	\$0.003028	1.00×	2026-08-28T14:43:55.703507+00:00
Study 2 · Gemini	structured_no_validation	150/150 (0 missing)	2.000	2129.6	0.95×	4.73	5.70	8.86	\$0.002612	0.86×	2026-08-28T14:43:55.703507+00:00
Study 2 · Gemini	contract	150/150 (0 missing)	2.007	2139.5	0.95×	4.76	5.58	9.69	\$0.002623	0.87×	2026-08-28T14:43:55.703507+00:00
Study 2 · Luna	baseline	150/150 (0 missing)	1.000	1296.8	0.53×	10.17	26.84	33.23	\$0.001289	0.63×	2026-08-28T14:43:55.703507+00:00
Study 2 · Luna	naive	150/150 (0 missing)	2.000	2456.1	1.00×	18.38	31.21	46.77	\$0.002051	1.00×	2026-08-28T14:43:55.703507+00:00
Study 2 · Luna	structured_no_validation	150/150 (0 missing)	2.000	3153.1	1.28×	20.52	38.53	51.06	\$0.002459	1.20×	2026-08-28T14:43:55.703507+00:00
Study 2 · Luna	contract	150/150 (0 missing)	2.000	3174.5	1.29×	20.22	37.59	47.67	\$0.002476	1.21×	2026-08-28T14:43:55.703507+00:00

Notes. Total tokens are input plus output tokens; reasoning and cached-token fields are not added again. The two Study 1A cells have no recorded pricing snapshot timestamp; their logged costs are displayed as logged, with no price-list reconstruction or imputation. P95 and median follow the existing analysis implementation: sorted observations with position $q^*(n-1)$ and linear interpolation.

Table S5. Error classes within evaluator regimes

Descriptive error-class counts within their original evaluator regimes. EvalPlus reports error_class and base_error_class as distinct blocks because they describe different harness fields. BigCodeBench-Hard reports only canonical error_class rows: its base_error_class records are exact compatibility aliases and are omitted rather than duplicated as a second scientific result. Every count retains the 150 arm-run denominator, including explicit zero-filled taxonomy rows. Rows are not pooled across models or evaluator regimes and do not estimate task success or complexity moderation.

Evaluator regime	Model-study cell	Arm	Evaluator field	Error class	Count / runs (share)	Zero-filled
evalplus_harness	Study 1A · Gemini	baseline	error_class	no_error	110/150 (73.3%)	no
evalplus_harness	Study 1A · Gemini	baseline	error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1A · Gemini	baseline	error_class	runtime	3/150 (2.0%)	no
evalplus_harness	Study 1A · Gemini	baseline	error_class	assertion	37/150 (24.7%)	no
evalplus_harness	Study 1A · Gemini	baseline	error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1A · Gemini	baseline	base_error_class	no_error	134/150 (89.3%)	no
evalplus_harness	Study 1A · Gemini	baseline	base_error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1A · Gemini	baseline	base_error_class	runtime	3/150 (2.0%)	no
evalplus_harness	Study 1A · Gemini	baseline	base_error_class	assertion	13/150 (8.7%)	no
evalplus_harness	Study 1A · Gemini	baseline	base_error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1A · Gemini	naive	error_class	no_error	112/150 (74.7%)	no
evalplus_harness	Study 1A · Gemini	naive	error_class	syntax	1/150 (0.7%)	no
evalplus_harness	Study 1A · Gemini	naive	error_class	runtime	8/150 (5.3%)	no
evalplus_harness	Study 1A · Gemini	naive	error_class	assertion	29/150 (19.3%)	no
evalplus_harness	Study 1A · Gemini	naive	error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1A · Gemini	naive	base_error_class	no_error	134/150 (89.3%)	no
evalplus_harness	Study 1A · Gemini	naive	base_error_class	syntax	1/150 (0.7%)	no
evalplus_harness	Study 1A · Gemini	naive	base_error_class	runtime	2/150 (1.3%)	no
evalplus_harness	Study 1A · Gemini	naive	base_error_class	assertion	13/150 (8.7%)	no
evalplus_harness	Study 1A · Gemini	naive	base_error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1A · Gemini	structured_no_validation	error_class	no_error	116/150 (77.3%)	no
evalplus_harness	Study 1A · Gemini	structured_no_validation	error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1A · Gemini	structured_no_validation	error_class	runtime	9/150 (6.0%)	no
evalplus_harness	Study 1A · Gemini	structured_no_validation	error_class	assertion	25/150 (16.7%)	no
evalplus_harness	Study 1A · Gemini	structured_no_validation	error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1A · Gemini	structured_no_validation	base_error_class	no_error	136/150 (90.7%)	no
evalplus_harness	Study 1A · Gemini	structured_no_validation	base_error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1A · Gemini	structured_no_validation	base_error_class	runtime	4/150 (2.7%)	no
evalplus_harness	Study 1A · Gemini	structured_no_validation	base_error_class	assertion	10/150 (6.7%)	no
evalplus_harness	Study 1A · Gemini	structured_no_validation	base_error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1A · Gemini	contract	error_class	no_error	114/150 (76.0%)	no
evalplus_harness	Study 1A · Gemini	contract	error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1A · Gemini	contract	error_class	runtime	8/150 (5.3%)	no
evalplus_harness	Study 1A · Gemini	contract	error_class	assertion	28/150 (18.7%)	no
evalplus_harness	Study 1A · Gemini	contract	error_class	timeout	0/150 (0.0%)	yes

Evaluator regime	Model-study cell	Arm	Evaluator field	Error class	Count / runs (share)	Zero-filled
evalplus_harness	Study 1A · Gemini	contract	base_error_class	no_error	137/150 (91.3%)	no
evalplus_harness	Study 1A · Gemini	contract	base_error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1A · Gemini	contract	base_error_class	runtime	3/150 (2.0%)	no
evalplus_harness	Study 1A · Gemini	contract	base_error_class	assertion	10/150 (6.7%)	no
evalplus_harness	Study 1A · Gemini	contract	base_error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	baseline	error_class	no_error	112/150 (74.7%)	no
evalplus_harness	Study 1B · Gemini	baseline	error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	baseline	error_class	runtime	1/150 (0.7%)	no
evalplus_harness	Study 1B · Gemini	baseline	error_class	assertion	37/150 (24.7%)	no
evalplus_harness	Study 1B · Gemini	baseline	error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	baseline	base_error_class	no_error	135/150 (90.0%)	no
evalplus_harness	Study 1B · Gemini	baseline	base_error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	baseline	base_error_class	runtime	1/150 (0.7%)	no
evalplus_harness	Study 1B · Gemini	baseline	base_error_class	assertion	14/150 (9.3%)	no
evalplus_harness	Study 1B · Gemini	baseline	base_error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	naive	error_class	no_error	113/150 (75.3%)	no
evalplus_harness	Study 1B · Gemini	naive	error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	naive	error_class	runtime	9/150 (6.0%)	no
evalplus_harness	Study 1B · Gemini	naive	error_class	assertion	28/150 (18.7%)	no
evalplus_harness	Study 1B · Gemini	naive	error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	naive	base_error_class	no_error	136/150 (90.7%)	no
evalplus_harness	Study 1B · Gemini	naive	base_error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	naive	base_error_class	runtime	3/150 (2.0%)	no
evalplus_harness	Study 1B · Gemini	naive	base_error_class	assertion	11/150 (7.3%)	no
evalplus_harness	Study 1B · Gemini	naive	base_error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	structured_no_validation	error_class	no_error	114/150 (76.0%)	no
evalplus_harness	Study 1B · Gemini	structured_no_validation	error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	structured_no_validation	error_class	runtime	9/150 (6.0%)	no
evalplus_harness	Study 1B · Gemini	structured_no_validation	error_class	assertion	27/150 (18.0%)	no
evalplus_harness	Study 1B · Gemini	structured_no_validation	error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	structured_no_validation	base_error_class	no_error	134/150 (89.3%)	no
evalplus_harness	Study 1B · Gemini	structured_no_validation	base_error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	structured_no_validation	base_error_class	runtime	3/150 (2.0%)	no
evalplus_harness	Study 1B · Gemini	structured_no_validation	base_error_class	assertion	13/150 (8.7%)	no
evalplus_harness	Study 1B · Gemini	structured_no_validation	base_error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	contract	error_class	no_error	115/150 (76.7%)	no
evalplus_harness	Study 1B · Gemini	contract	error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	contract	error_class	runtime	7/150 (4.7%)	no
evalplus_harness	Study 1B · Gemini	contract	error_class	assertion	28/150 (18.7%)	no

Evaluator regime	Model-study cell	Arm	Evaluator field	Error class	Count / runs (share)	Zero-filled
evalplus_harness	Study 1B · Gemini	contract	error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	contract	base_error_class	no_error	136/150 (90.7%)	no
evalplus_harness	Study 1B · Gemini	contract	base_error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Gemini	contract	base_error_class	runtime	3/150 (2.0%)	no
evalplus_harness	Study 1B · Gemini	contract	base_error_class	assertion	11/150 (7.3%)	no
evalplus_harness	Study 1B · Gemini	contract	base_error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	baseline	error_class	no_error	118/150 (78.7%)	no
evalplus_harness	Study 1B · Luna	baseline	error_class	syntax	1/150 (0.7%)	no
evalplus_harness	Study 1B · Luna	baseline	error_class	runtime	10/150 (6.7%)	no
evalplus_harness	Study 1B · Luna	baseline	error_class	assertion	21/150 (14.0%)	no
evalplus_harness	Study 1B · Luna	baseline	error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	baseline	base_error_class	no_error	142/150 (94.7%)	no
evalplus_harness	Study 1B · Luna	baseline	base_error_class	syntax	1/150 (0.7%)	no
evalplus_harness	Study 1B · Luna	baseline	base_error_class	runtime	4/150 (2.7%)	no
evalplus_harness	Study 1B · Luna	baseline	base_error_class	assertion	3/150 (2.0%)	no
evalplus_harness	Study 1B · Luna	baseline	base_error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	naive	error_class	no_error	106/150 (70.7%)	no
evalplus_harness	Study 1B · Luna	naive	error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	naive	error_class	runtime	16/150 (10.7%)	no
evalplus_harness	Study 1B · Luna	naive	error_class	assertion	28/150 (18.7%)	no
evalplus_harness	Study 1B · Luna	naive	error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	naive	base_error_class	no_error	143/150 (95.3%)	no
evalplus_harness	Study 1B · Luna	naive	base_error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	naive	base_error_class	runtime	3/150 (2.0%)	no
evalplus_harness	Study 1B · Luna	naive	base_error_class	assertion	4/150 (2.7%)	no
evalplus_harness	Study 1B · Luna	naive	base_error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	structured_no_validation	error_class	no_error	110/150 (73.3%)	no
evalplus_harness	Study 1B · Luna	structured_no_validation	error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	structured_no_validation	error_class	runtime	17/150 (11.3%)	no
evalplus_harness	Study 1B · Luna	structured_no_validation	error_class	assertion	23/150 (15.3%)	no
evalplus_harness	Study 1B · Luna	structured_no_validation	error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	structured_no_validation	base_error_class	no_error	142/150 (94.7%)	no
evalplus_harness	Study 1B · Luna	structured_no_validation	base_error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	structured_no_validation	base_error_class	runtime	3/150 (2.0%)	no
evalplus_harness	Study 1B · Luna	structured_no_validation	base_error_class	assertion	5/150 (3.3%)	no
evalplus_harness	Study 1B · Luna	structured_no_validation	base_error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	contract	error_class	no_error	112/150 (74.7%)	no
evalplus_harness	Study 1B · Luna	contract	error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	contract	error_class	runtime	16/150 (10.7%)	no

Evaluator regime	Model-study cell	Arm	Evaluator field	Error class	Count / runs (share)	Zero-filled
evalplus_harness	Study 1B · Luna	contract	error_class	assertion	22/150 (14.7%)	no
evalplus_harness	Study 1B · Luna	contract	error_class	timeout	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	contract	base_error_class	no_error	143/150 (95.3%)	no
evalplus_harness	Study 1B · Luna	contract	base_error_class	syntax	0/150 (0.0%)	yes
evalplus_harness	Study 1B · Luna	contract	base_error_class	runtime	3/150 (2.0%)	no
evalplus_harness	Study 1B · Luna	contract	base_error_class	assertion	4/150 (2.7%)	no
evalplus_harness	Study 1B · Luna	contract	base_error_class	timeout	0/150 (0.0%)	yes
bigcodebench_hidden_test	Study 2 · Gemini	baseline	error_class	no_error	58/150 (38.7%)	no
bigcodebench_hidden_test	Study 2 · Gemini	baseline	error_class	assertion	92/150 (61.3%)	no
bigcodebench_hidden_test	Study 2 · Gemini	baseline	error_class	timeout	0/150 (0.0%)	yes
bigcodebench_hidden_test	Study 2 · Gemini	naive	error_class	no_error	51/150 (34.0%)	no
bigcodebench_hidden_test	Study 2 · Gemini	naive	error_class	assertion	99/150 (66.0%)	no
bigcodebench_hidden_test	Study 2 · Gemini	naive	error_class	timeout	0/150 (0.0%)	yes
bigcodebench_hidden_test	Study 2 · Gemini	structured_no_validation	error_class	no_error	56/150 (37.3%)	no
bigcodebench_hidden_test	Study 2 · Gemini	structured_no_validation	error_class	assertion	94/150 (62.7%)	no
bigcodebench_hidden_test	Study 2 · Gemini	structured_no_validation	error_class	timeout	0/150 (0.0%)	yes
bigcodebench_hidden_test	Study 2 · Gemini	contract	error_class	no_error	56/150 (37.3%)	no
bigcodebench_hidden_test	Study 2 · Gemini	contract	error_class	assertion	94/150 (62.7%)	no
bigcodebench_hidden_test	Study 2 · Gemini	contract	error_class	timeout	0/150 (0.0%)	yes
bigcodebench_hidden_test	Study 2 · Luna	baseline	error_class	no_error	59/150 (39.3%)	no
bigcodebench_hidden_test	Study 2 · Luna	baseline	error_class	assertion	91/150 (60.7%)	no
bigcodebench_hidden_test	Study 2 · Luna	baseline	error_class	timeout	0/150 (0.0%)	yes
bigcodebench_hidden_test	Study 2 · Luna	naive	error_class	no_error	46/150 (30.7%)	no
bigcodebench_hidden_test	Study 2 · Luna	naive	error_class	assertion	103/150 (68.7%)	no
bigcodebench_hidden_test	Study 2 · Luna	naive	error_class	timeout	1/150 (0.7%)	no
bigcodebench_hidden_test	Study 2 · Luna	structured_no_validation	error_class	no_error	47/150 (31.3%)	no
bigcodebench_hidden_test	Study 2 · Luna	structured_no_validation	error_class	assertion	103/150 (68.7%)	no
bigcodebench_hidden_test	Study 2 · Luna	structured_no_validation	error_class	timeout	0/150 (0.0%)	yes
bigcodebench_hidden_test	Study 2 · Luna	contract	error_class	no_error	48/150 (32.0%)	no
bigcodebench_hidden_test	Study 2 · Luna	contract	error_class	assertion	102/150 (68.0%)	no
bigcodebench_hidden_test	Study 2 · Luna	contract	error_class	timeout	0/150 (0.0%)	yes

Notes. Coverage is the preregistered five-branch F8 error-class package. Study 1A - DeepSeek is explicitly outside this source scope (study1a_deepseek_in_scope=false); no raw reconstruction is attempted. Zero-filled means the class was part of the fixed regime taxonomy and was observed zero times; it does not mean missing data. The 24 omitted Study 2 base_error_class rows plus the 144 displayed rows account for all 168 source rows.

Table S6. Full study-design and analysis-scope inventory

Six model–study cells across three studies. Every cell runs the same four-arm design (baseline, naive, structured_no_validation, contract) over 50 tasks with 3 repetitions per task, giving 600 executions per cell and 150 per arm. Study 1A and Study 1B draw on the same EvalPlus held-out task sample; Study 2 uses a separate BigCodeBench-Hard set. Success inference is clustered on the task, so each cell contributes 50 independent units rather than 600, and cells are never pooled across models or task regimes. A further study (Study 3) was deferred before data collection (S3-NOT-RUN) and contributes no executions, estimands, or rows.

Model–study cell	Role in this paper	Benchmark (task set)	Tasks × repeats × arms	Executions per cell	Executions per arm	Inference unit
Study 1A · Gemini	Primary/original study	EvalPlus held-out (tasks_heldout)	$50 \times 3 \times 4$	600	150	task
Study 1A · DeepSeek	Cross-model replication of the primary study	EvalPlus held-out (tasks_heldout)	$50 \times 3 \times 4$	600	150	task
Study 1B · Gemini	Model-family extension and partial replication	EvalPlus held-out (tasks_heldout)	$50 \times 3 \times 4$	600	150	task
Study 1B · Luna	Model-family extension and partial replication	EvalPlus held-out (tasks_heldout)	$50 \times 3 \times 4$	600	150	task
Study 2 · Gemini	Generalization to a harder benchmark (primary confirmatory analysis)	BigCodeBench-Hard (tasks_study2_complex)	$50 \times 3 \times 4$	600	150	task
Study 2 · Luna	Pre-specified replication on a harder benchmark	BigCodeBench-Hard (tasks_study2_complex)	$50 \times 3 \times 4$	600	150	task

Notes. Task count, repetitions and arm count are identical across cells by design; they are repeated on every row so that each denominator sits next to the cell it belongs to. The inference unit named here is that of the task-success estimand in main-text Table 1 and in the mechanism figure. Main-text Tables 2 and 3 are descriptive summaries whose unit is the arm-run; the two units are not interchangeable and are never added together. Study 1B reuses Study 1A's task sample, so it is a model-family extension and partial replication rather than an independent replication on new tasks.

Table S7. Study 1A MAST failure-mode composition and arm incidence

Primary MAST failure mode for every decided Study 1A failure, by model-study cell and arm. Two different denominators are reported side by side and must not be conflated. Composition divides the mode count by the decided failures of that same arm and answers what failures looked like once they happened; incidence divides the same count by the 150 completed arm-runs of that arm and answers how often that mode was seen at all. Within each cell and arm the mode counts sum exactly to the decided failures of that arm. Only failed arm-runs were sent to the labeling pipeline; successful runs were never labeled, so no mode column describes successful executions. Rows with a count of zero are kept: the mode was available to the labelers and no record received it, which is a measured zero rather than a missing observation. Records whose decided label was none are kept as their own row and are not folded into another mode. A MAST mode describes how the multi-agent interaction failed and is not an evaluator error class; the two taxonomies live in different tables and are never mapped onto each other. Labeling is blind to producer model identity and to the self label but not to the arm. This table is descriptive, is not a main contribution of the paper, and supports no causal reading. Gemini and DeepSeek are separate experiments and are never pooled.

Model-study cell	Arm	Primary MAST failure mode	Records with this mode	Decided failures in arm (denominator 1)	Composition among decided failures	Completed arm-runs in arm (denominator 2)	Incidence among completed arm-runs
Study 1A · Gemini	Baseline	FM-1.1 Disobey Task Specification	31	40	77.5%	150	20.7%
Study 1A · Gemini	Baseline	FM-1.5 Unaware of Termination Conditions	0	40	0.0%	150	0.0%
Study 1A · Gemini	Baseline	FM-2.2 Fail to Ask for Clarification	0	40	0.0%	150	0.0%
Study 1A · Gemini	Baseline	FM-2.5 Ignored Other Agent's Input	0	40	0.0%	150	0.0%
Study 1A · Gemini	Baseline	FM-2.6 Action-Reasoning Mismatch	0	40	0.0%	150	0.0%
Study 1A · Gemini	Baseline	FM-3.1 Premature Termination	1	40	2.5%	150	0.7%
Study 1A · Gemini	Baseline	FM-3.2 Weak Verification	6	40	15.0%	150	4.0%
Study 1A · Gemini	Baseline	FM-3.3 No or Incorrect Verification	2	40	5.0%	150	1.3%
Study 1A · Gemini	Naive multi-agent	FM-1.1 Disobey Task Specification	30	38	78.9%	150	20.0%
Study 1A · Gemini	Naive multi-agent	FM-1.5 Unaware of Termination Conditions	0	38	0.0%	150	0.0%
Study 1A · Gemini	Naive multi-agent	FM-2.2 Fail to Ask for Clarification	5	38	13.2%	150	3.3%
Study 1A · Gemini	Naive multi-agent	FM-2.5 Ignored Other Agent's Input	0	38	0.0%	150	0.0%
Study 1A · Gemini	Naive multi-agent	FM-2.6 Action-Reasoning Mismatch	0	38	0.0%	150	0.0%
Study 1A · Gemini	Naive multi-agent	FM-3.1 Premature Termination	0	38	0.0%	150	0.0%
Study 1A · Gemini	Naive multi-agent	FM-3.2 Weak Verification	1	38	2.6%	150	0.7%
Study 1A · Gemini	Naive multi-agent	FM-3.3 No or Incorrect Verification	2	38	5.3%	150	1.3%
Study 1A · Gemini	Structured (no validation)	FM-1.1 Disobey Task Specification	19	34	55.9%	150	12.7%
Study 1A · Gemini	Structured (no validation)	FM-1.5 Unaware of Termination Conditions	0	34	0.0%	150	0.0%
Study 1A · Gemini	Structured (no validation)	FM-2.2 Fail to Ask for Clarification	1	34	2.9%	150	0.7%
Study 1A · Gemini	Structured (no validation)	FM-2.5 Ignored Other Agent's Input	3	34	8.8%	150	2.0%
Study 1A · Gemini	Structured (no validation)	FM-2.6 Action-Reasoning Mismatch	4	34	11.8%	150	2.7%
Study 1A · Gemini	Structured (no validation)	FM-3.1 Premature Termination	6	34	17.6%	150	4.0%

Model-study cell	Arm	Primary MAST failure mode	Records with this mode	Decided failures in arm (denominator 1)	Composition among decided failures	Completed arm-runs in arm (denominator 2)	Incidence among completed arm-runs
Study 1A · Gemini	Structured (no validation)	FM-3.2 Weak Verification	1	34	2.9%	150	0.7%
Study 1A · Gemini	Structured (no validation)	FM-3.3 No or Incorrect Verification	0	34	0.0%	150	0.0%
Study 1A · Gemini	Contract	FM-1.1 Disobey Task Specification	23	36	63.9%	150	15.3%
Study 1A · Gemini	Contract	FM-1.5 Unaware of Termination Conditions	1	36	2.8%	150	0.7%
Study 1A · Gemini	Contract	FM-2.2 Fail to Ask for Clarification	0	36	0.0%	150	0.0%
Study 1A · Gemini	Contract	FM-2.5 Ignored Other Agent's Input	1	36	2.8%	150	0.7%
Study 1A · Gemini	Contract	FM-2.6 Action-Reasoning Mismatch	0	36	0.0%	150	0.0%
Study 1A · Gemini	Contract	FM-3.1 Premature Termination	4	36	11.1%	150	2.7%
Study 1A · Gemini	Contract	FM-3.2 Weak Verification	5	36	13.9%	150	3.3%
Study 1A · Gemini	Contract	FM-3.3 No or Incorrect Verification	2	36	5.6%	150	1.3%
Study 1A · DeepSeek	Baseline	FM-1.1 Disobey Task Specification	22	33	66.7%	150	14.7%
Study 1A · DeepSeek	Baseline	FM-2.2 Fail to Ask for Clarification	0	33	0.0%	150	0.0%
Study 1A · DeepSeek	Baseline	FM-2.6 Action-Reasoning Mismatch	0	33	0.0%	150	0.0%
Study 1A · DeepSeek	Baseline	FM-3.1 Premature Termination	0	33	0.0%	150	0.0%
Study 1A · DeepSeek	Baseline	FM-3.2 Weak Verification	7	33	21.2%	150	4.7%
Study 1A · DeepSeek	Baseline	FM-3.3 No or Incorrect Verification	3	33	9.1%	150	2.0%
Study 1A · DeepSeek	Baseline	none (decided, but no failure mode assigned)	1	33	3.0%	150	0.7%
Study 1A · DeepSeek	Naive multi-agent	FM-1.1 Disobey Task Specification	29	43	67.4%	150	19.3%
Study 1A · DeepSeek	Naive multi-agent	FM-2.2 Fail to Ask for Clarification	3	43	7.0%	150	2.0%
Study 1A · DeepSeek	Naive multi-agent	FM-2.6 Action-Reasoning Mismatch	0	43	0.0%	150	0.0%
Study 1A · DeepSeek	Naive multi-agent	FM-3.1 Premature Termination	6	43	14.0%	150	4.0%
Study 1A · DeepSeek	Naive multi-agent	FM-3.2 Weak Verification	3	43	7.0%	150	2.0%
Study 1A · DeepSeek	Naive multi-agent	FM-3.3 No or Incorrect Verification	2	43	4.7%	150	1.3%
Study 1A · DeepSeek	Naive multi-agent	none (decided, but no failure mode assigned)	0	43	0.0%	150	0.0%
Study 1A · DeepSeek	Structured (no validation)	FM-1.1 Disobey Task Specification	26	38	68.4%	150	17.3%
Study 1A · DeepSeek	Structured (no validation)	FM-2.2 Fail to Ask for Clarification	4	38	10.5%	150	2.7%
Study 1A · DeepSeek	Structured (no validation)	FM-2.6 Action-Reasoning	1	38	2.6%	150	0.7%

Model-study cell	Arm	Primary MAST failure mode	Records with this mode	Decided failures in arm (denominator 1)	Composition among decided failures	Completed arm-runs in arm (denominator 2)	Incidence among completed arm-runs
		Mismatch					
Study 1A · DeepSeek	Structured (no validation)	FM-3.1 Premature Termination	2	38	5.3%	150	1.3%
Study 1A · DeepSeek	Structured (no validation)	FM-3.2 Weak Verification	0	38	0.0%	150	0.0%
Study 1A · DeepSeek	Structured (no validation)	FM-3.3 No or Incorrect Verification	2	38	5.3%	150	1.3%
Study 1A · DeepSeek	Structured (no validation)	none (decided, but no failure mode assigned)	3	38	7.9%	150	2.0%
Study 1A · DeepSeek	Contract	FM-1.1 Disobey Task Specification	24	39	61.5%	150	16.0%
Study 1A · DeepSeek	Contract	FM-2.2 Fail to Ask for Clarification	2	39	5.1%	150	1.3%
Study 1A · DeepSeek	Contract	FM-2.6 Action-Reasoning Mismatch	0	39	0.0%	150	0.0%
Study 1A · DeepSeek	Contract	FM-3.1 Premature Termination	1	39	2.6%	150	0.7%
Study 1A · DeepSeek	Contract	FM-3.2 Weak Verification	7	39	17.9%	150	4.7%
Study 1A · DeepSeek	Contract	FM-3.3 No or Incorrect Verification	4	39	10.3%	150	2.7%
Study 1A · DeepSeek	Contract	none (decided, but no failure mode assigned)	1	39	2.6%	150	0.7%

Notes. Composition and incidence are recomputed from the printed count and are accepted only when they agree with the value recorded in the source row at that source's own precision. This file stores both ratios rounded to six decimal places, so the comparison is made at six places; sources that store a ratio at full precision are compared without that rounding. The mode vocabulary differs between the two cells because each panel only ever emitted the modes listed here; the vocabulary is not padded to a common grid. Mode names come from the frozen MAST taxonomy that the judges themselves were shown, so a printed name cannot drift from the definition used at labeling time. A decided failure is an arm-run that failed and reached a final label; the two external judges decide the label and the producer model's own self-label is diagnostic only.

Table S8. Study 1A human validation of the MAST labeling pipeline

Descriptive agreement measures from the blind human round, reported separately for each Study 1A cell and never pooled. Two annotators labeled from the evidence package alone and did not see any AI label while labeling. After the two blind label sets were locked, the adjudication stage produced one final human label for every reviewed record; where the two annotators differed that difference was an input to adjudication, but the stage was not restricted to those records. Each metric keeps its own comparable-record denominator, which is why the denominators differ down the column: the human-to-human and kappa rows cover the whole reviewed sample, while the two adjudication rows cover the disjoint subsets where the external AI judges agreed and where they split. Those two subsets sum exactly to the reviewed sample, so the adjudication accounting is complete. Cohen kappa is a chance-corrected coefficient and is printed as a coefficient; it is deliberately not converted to a percentage, because doing so would invite reading it as raw agreement. The reviewed records are a selected, disagreement-enriched sample and not a random draw. The agreement and kappa values are therefore descriptive properties of this sample alone; how the sample was selected can affect how they should be read, and they do not generalize to the whole error population. This table validates the labeling pipeline; it is not evidence about task success and supports no causal reading.

Model-study cell	Human-reviewed sample size	Metric	Reported value	Comparable records for this metric	Sampling and selection status
Study 1A · Gemini	96	Blind human-to-human exact agreement	26.0%	96	Selected, disagreement-enriched; not a random sample
Study 1A · Gemini	96	Cohen kappa (coefficient, not a percentage)	0.026	96	Selected, disagreement-enriched; not a random sample
Study 1A · Gemini	96	Final human label vs external AI consensus	55.6%	27	Selected, disagreement-enriched; not a random sample
Study 1A · Gemini	96	Final human label vs adjudicator on external split	21.7%	69	Selected, disagreement-enriched; not a random sample
Study 1A · DeepSeek	103	Blind human-to-human exact agreement	67.0%	103	Selected, disagreement-enriched; not a random sample
Study 1A · DeepSeek	103	Cohen kappa (coefficient, not a percentage)	0.0107	103	Selected, disagreement-enriched; not a random sample
Study 1A · DeepSeek	103	Final human label vs external AI consensus	29.4%	17	Selected, disagreement-enriched; not a random sample
Study 1A · DeepSeek	103	Final human label vs adjudicator on external split	25.6%	86	Selected, disagreement-enriched; not a random sample

Notes. The two adjudication rows partition the reviewed sample: 27 plus 69 equals 96 for Gemini and 17 plus 86 equals 103 for DeepSeek. Both rows compare the final human label, so the partition is also what shows that a final human label exists for every reviewed record and not only for the records where the external judges split. The partition is verified against the source before the table is written. Agreement and kappa are reported as descriptive properties of the reviewed sample and not as estimates of labeling reliability over all failures. This table does not carry the marginal label distributions that any account of why a coefficient takes its value would require, and none is offered here. Human annotators saw the same blind evidence package as the AI judges: producer model identity and the self label were hidden, the arm was not. Percentages are recomputed from the stored proportion; no value is rounded twice.

Table S9 Panel A. Study 1A exploratory self-consistency and Plus failure

Panel A of Table S9. Rank association between a task's self-consistency agreement and its observed Plus failure rate, for the two Study 1A producer models kept in separate rows and never pooled into a shared coefficient. The baseline arm is the primary pairing and the reason is structural rather than chosen after the fact: the self-consistency candidates are generated from the baseline prompt with no planner and no contract layer, so the baseline arm is the pairing that matches how the candidates were produced. The remaining five pairings are secondary and descriptive. This panel is exploratory in the sense fixed before the data were seen: no significance test, p-value or confidence interval is computed or attached, and no threshold is applied to any coefficient. The sign of a coefficient is not a prediction of task outcome, does not define a screening rule, and supports no causal reading. The result is confined to Study 1A and does not extend to Study 1B or Study 2, which used different regimes and models. Agreement is a provider-independent proxy: candidates are drawn at temperature 0.8 while the arms run at the frozen default, so this is not a direct measurement of the baseline arm's own sampling distribution. Coefficients are printed at the precision recorded in the source and are not rounded for display.

Model-study cell	Pairing	Role	Tasks	Spearman rho	Tie handling	Interpretation status
Study 1A · Gemini	Baseline arm	Primary	50	-0.505254	average_rank_pearson_v1	Exploratory only: no test, no threshold, no prediction
Study 1A · Gemini	Contract arm	Secondary	50	-0.385867	average_rank_pearson_v1	Exploratory only: no test, no threshold, no prediction
Study 1A · Gemini	Naive arm	Secondary	50	-0.463967	average_rank_pearson_v1	Exploratory only: no test, no threshold, no prediction
Study 1A · Gemini	All twelve arm-runs of the task	Secondary	50	-0.45163	average_rank_pearson_v1	Exploratory only: no test, no threshold, no prediction
Study 1A · Gemini	Self-consistency round, own oracle	Secondary	50	-0.470329	average_rank_pearson_v1	Exploratory only: no test, no threshold, no prediction
Study 1A · Gemini	Structured arm (no validation)	Secondary	50	-0.276411	average_rank_pearson_v1	Exploratory only: no test, no threshold, no prediction
Study 1A · DeepSeek	Baseline arm	Primary	50	-0.330661	average_rank_pearson_v1	Exploratory only: no test, no threshold, no prediction
Study 1A · DeepSeek	Contract arm	Secondary	50	-0.247828	average_rank_pearson_v1	Exploratory only: no test, no threshold, no prediction
Study 1A · DeepSeek	Naive arm	Secondary	50	-0.075095	average_rank_pearson_v1	Exploratory only: no test, no threshold, no prediction
Study 1A · DeepSeek	All twelve arm-runs of the task	Secondary	50	-0.206312	average_rank_pearson_v1	Exploratory only: no test, no threshold, no prediction
Study 1A · DeepSeek	Self-consistency round, own oracle	Secondary	50	-0.404258	average_rank_pearson_v1	Exploratory only: no test, no threshold, no prediction
Study 1A · DeepSeek	Structured arm (no validation)	Secondary	50	-0.292527	average_rank_pearson_v1	Exploratory only: no test, no threshold, no prediction

Notes. Twelve source rows produce twelve table rows in the same order; nothing is aggregated, so every coefficient traces to one row. Ties are resolved by the recorded method and the method is printed in its own column rather than left to a caption. The mean self-consistency agreement figure is deliberately absent: it was closed as not required by F9 and is not reintroduced here. Figure S5 draws these same twelve coefficients; both are built directly from the C8 source rather than from each other.

Table S9 Panel B. Base-to-Plus attrition by arm and evaluator regime

Panel B of Table S9. How many runs that passed the Base tests then failed the stricter Plus tests, per arm, with every model-study cell kept separate and no arm or cell pooled. The denominator of the attrition column is the Base-pass runs of that row, not the 150 completed arm-runs: the quantity answers how often a run that already worked stopped working under the hidden tests. In the EvalPlus held-out regime Base and Plus are two evaluations of the same run, so the loss is measurable and is reported. In the BigCodeBench-Hard regime of Study 2 they are not: `plus_pass` is a compatibility alias of `base_pass`, the two counts are one evaluation reported twice, and there is no second hidden-test evaluation for a run to be lost to. Those eight rows therefore print NA with the reason in the cell itself. NA here means the quantity was never measurable, not that it was measured and came out at zero, and the table refuses to print a zero or the phrase no attrition for them. For the same reason a Study 2 row must not be read as evidence of hidden-test robustness: no hidden-test comparison exists there. The panel is descriptive and supports no causal reading; the arms are not equal-compute conditions and the baseline arm in particular issues a single model call per task.

Model-study cell	Evaluator regime	Arm	Completed arm-runs	Base-pass runs	Plus-pass runs	Base-pass and Plus-fail runs	Attrition among Base-pass runs	Base/Plus separability status
Study 1A · Gemini	EvalPlus held-out	Baseline	150	134	110	24	17.9%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1A · Gemini	EvalPlus held-out	Naive multi-agent	150	134	112	22	16.4%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1A · Gemini	EvalPlus held-out	Structured (no validation)	150	136	116	20	14.7%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1A · Gemini	EvalPlus held-out	Contract	150	137	114	23	16.8%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1A · DeepSeek	EvalPlus held-out	Baseline	150	137	117	20	14.6%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1A · DeepSeek	EvalPlus held-out	Naive multi-agent	150	137	107	30	21.9%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1A · DeepSeek	EvalPlus held-out	Structured (no validation)	150	141	112	29	20.6%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1A · DeepSeek	EvalPlus held-out	Contract	150	138	111	27	19.6%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1B · Gemini	EvalPlus held-out	Baseline	150	135	112	23	17.0%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1B · Gemini	EvalPlus held-out	Naive multi-agent	150	136	113	23	16.9%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1B · Gemini	EvalPlus held-out	Structured (no validation)	150	134	114	20	14.9%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1B · Gemini	EvalPlus held-out	Contract	150	136	115	21	15.4%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1B · Luna	EvalPlus held-out	Baseline	150	142	118	24	16.9%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1B · Luna	EvalPlus held-out	Naive multi-agent	150	143	106	37	25.9%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1B · Luna	EvalPlus held-out	Structured (no validation)	150	142	110	32	22.5%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus
Study 1B · Luna	EvalPlus held-out	Contract	150	143	112	31	21.7%	Separable: Base and Plus are two evaluations of the same run, so a run can pass Base and fail Plus

Model-study cell	Evaluator regime	Arm	Completed arm-runs	Base-pass runs	Plus-pass runs	Base-pass and Plus-fail runs	Attrition among Base-pass runs	Base/Plus separability status
Study 2 · Gemini	BigCodeBench-Hard	Baseline	150	58	58	NA (not separable)	NA — Base and Plus are the same compatibility-aliased outcome; no separate hidden-test attrition measurement	Not separable: plus_pass is a compatibility alias of base_pass, so the two counts are one evaluation reported twice, not two independent evaluations
Study 2 · Gemini	BigCodeBench-Hard	Naive multi-agent	150	51	51	NA (not separable)	NA — Base and Plus are the same compatibility-aliased outcome; no separate hidden-test attrition measurement	Not separable: plus_pass is a compatibility alias of base_pass, so the two counts are one evaluation reported twice, not two independent evaluations
Study 2 · Gemini	BigCodeBench-Hard	Structured (no validation)	150	56	56	NA (not separable)	NA — Base and Plus are the same compatibility-aliased outcome; no separate hidden-test attrition measurement	Not separable: plus_pass is a compatibility alias of base_pass, so the two counts are one evaluation reported twice, not two independent evaluations
Study 2 · Gemini	BigCodeBench-Hard	Contract	150	56	56	NA (not separable)	NA — Base and Plus are the same compatibility-aliased outcome; no separate hidden-test attrition measurement	Not separable: plus_pass is a compatibility alias of base_pass, so the two counts are one evaluation reported twice, not two independent evaluations
Study 2 · Luna	BigCodeBench-Hard	Baseline	150	59	59	NA (not separable)	NA — Base and Plus are the same compatibility-aliased outcome; no separate hidden-test attrition measurement	Not separable: plus_pass is a compatibility alias of base_pass, so the two counts are one evaluation reported twice, not two independent evaluations
Study 2 · Luna	BigCodeBench-Hard	Naive multi-agent	150	46	46	NA (not separable)	NA — Base and Plus are the same compatibility-aliased outcome; no separate hidden-test attrition measurement	Not separable: plus_pass is a compatibility alias of base_pass, so the two counts are one evaluation reported twice, not two independent evaluations
Study 2 · Luna	BigCodeBench-Hard	Structured (no validation)	150	47	47	NA (not separable)	NA — Base and Plus are the same compatibility-aliased outcome; no separate hidden-test attrition measurement	Not separable: plus_pass is a compatibility alias of base_pass, so the two counts are one evaluation reported twice, not two independent evaluations
Study 2 · Luna	BigCodeBench-Hard	Contract	150	48	48	NA (not separable)	NA — Base and Plus are the same compatibility-aliased outcome; no separate hidden-test attrition measurement	Not separable: plus_pass is a compatibility alias of base_pass, so the two counts are one evaluation reported twice, not two independent evaluations

Notes. Base-pass and Plus-fail counts and attrition rates are recomputed from the printed counts and are accepted only when they match the source: Base-pass minus Plus-pass for the count, and that count divided by Base-pass runs for the rate. The separability of each row is read from the source flags rather than inferred from the regime, and the two are cross-checked against each other before the panel is written. Sixteen EvalPlus rows carry a measured attrition; eight Study 2 rows carry NA. No row is dropped from the audit for being NA. Every row covers 150 completed arm-runs, so the Base-pass column and the attrition denominator differ from row to row while the run count does not.

Table S10 Panel A. EvalPlus selection funnel for the shared Study 1 task sample

Panel A of Table S10. Deterministic construction of the 50-task EvalPlus-derived equality-compatible held-out subset shared by Studies 1A and 1B. HumanEval+ and MBPP+ are shown separately so that exclusion, eligibility and sampling counts retain their own denominators. The selected counts sum to 50; the eligible counts are larger because eligibility preceded deterministic sampling. This is task-set provenance, not a performance result.

Source dataset and version	Records examined	Excluded	Eligible	Selected	Distinct rejection reasons	Selection seed	Use in study program
HumanEval+ v0.1.10	164	34	130	30	8	20260727	Shared held-out sample for Studies 1A and 1B
MBPP+ v0.2.0	378	60	318	20	10	20260727	Shared held-out sample for Studies 1A and 1B

Notes. For each source dataset, excluded plus eligible equals examined, and selected never exceeds eligible; all identities are checked against the 50 hashes in the frozen selection manifest. The two studies reuse this same task sample. Study 1B is therefore a partial replication and model-family extension, not a replication on an independently sampled task set. The official description is EvalPlus-derived equality-compatible held-out subset; the numbers are not directly comparable with an EvalPlus leaderboard score.

Table S10 Panel B. Study 2 task partition and evaluator provenance

Panel B of Table S10. The 66-task BigCodeBench-Hard universe is partitioned into 16 development and 50 held-out tasks with no overlap; only the held-out partition enters the main statistics. Both partitions retain the same evaluation backend, metric, image identity and benchmark prompt field. The frozen parser package version did not match the observed package version, so package-version equality is not presented as the binding gate: exact content equality for the 66 prompts and payload identities was the binding evidence. This panel is provenance, not model performance.

Partition role	Task set	Tasks	In main statistics	Evaluation backend	Evaluation metric	Evaluator image identity	Benchmark prompt field	Parser binding status
Development	tasks_followup_dev	16	No	bigcodebench_untrusted_check_v1	bigcodebench_hidden_test_pass	sha256:3d80a30b7f8d2c511032afec5305bfa05f4e6e96db98459efc187814dd11c920	instruct_prompt	Content-bound; parser version pin not satisfied
Held-out	tasks_study2_complex	50	Yes	bigcodebench_untrusted_check_v1	bigcodebench_hidden_test_pass	sha256:3d80a30b7f8d2c511032afec5305bfa05f4e6e96db98459efc187814dd11c920	instruct_prompt	Content-bound; parser version pin not satisfied

Notes. The development and held-out counts sum to the 66-task source universe and the source manifests mark the partitions as disjoint. A false parser-pin field is not rewritten as a pass. It records a version mismatch; the content-identity gate is stated separately. Study 2 has one hidden-test outcome. Its compatibility fields named base_pass and plus_pass are aliases, not two evaluator layers.

Table S10 Panel C. Execution and evaluation provenance by model-study cell

Panel C of Table S10. Exact model/API routes, task regimes, prompt-configuration source manifests, evaluator identities and schema versions for all six model-study cells. A prompt hash is not copied out of an untracked run manifest: instead, the table identifies the per-cell run manifest whose byte hash is frozen in the tracked evidence manifest. This preserves the audit trail without claiming that a value absent from the publication package was independently recovered. Cells remain separate; this panel contains no outcomes.

Model-study cell	Model/API route	Task regime	Prompt/config provenance source	Evaluator provenance	Recorded schema versions	Experiment commit
Study 1A · Gemini	openrouter/google/gemini-3.5-flash-lite	evalplus_heldout	Per-cell run manifest; SHA-256 9ad753c583d60406...	EvalPlus Base/Plus; humanevalplus v0.1.10, mbppplus v0.2.0; metric plus_pass	result 2.0; analysis 2.3	91e7910bd13a
Study 1A · DeepSeek	openrouter/deepseek/deepseek-v4-flash	evalplus_heldout	Per-cell run manifest; SHA-256 4df3ab6ee304c2eb...	EvalPlus Base/Plus; humanevalplus v0.1.10, mbppplus v0.2.0; metric plus_pass	result 2.0; analysis 2.3	7bef753d0e94
Study 1B · Gemini	openrouter/google/gemini-3.5-flash-lite	evalplus_heldout	Per-cell run manifest; SHA-256 b0fc66e705f7c24b...	EvalPlus Base/Plus; humanevalplus v0.1.10, mbppplus v0.2.0; metric plus_pass	result 2.0; analysis 2.3	0fb04e4f2597
Study 1B · Luna	openrouter/openai/gpt-5.6-luna	evalplus_heldout	Per-cell run manifest; SHA-256 686f16b77715c01b...	EvalPlus Base/Plus; humanevalplus v0.1.10, mbppplus v0.2.0; metric plus_pass	result 2.0; analysis 2.3	4e58e592800b
Study 2 · Gemini	openrouter/google/gemini-3.5-flash-lite	bigcodebench_hard	Per-cell run manifest; SHA-256 e91dae3cc40d15d7...	bigcodebench_untrusted_check_v1; sha256:3d80a30b7f8d2c511032afec5305bfa05f4e6e96db98459efc187814dd11c920; one hidden-test outcome	result 2.0; analysis 2.3	58e5e2478fd9
Study 2 · Luna	openrouter/openai/gpt-5.6-luna	bigcodebench_hard	Per-cell run manifest; SHA-256 8636066980325631...	bigcodebench_untrusted_check_v1; sha256:3d80a30b7f8d2c511032afec5305bfa05f4e6e96db98459efc187814dd11c920; one hidden-test outcome	result 2.0; analysis 2.3	58e5e2478fd9

Notes. The run-manifest SHA-256 and full experiment commit are retained in metadata even though the visible cells shorten them for readability. EvalPlus cells have separate Base and Plus evaluations. Study 2 has one BigCodeBench-Hard hidden-test outcome and no Base-to-Plus layer. Internal analysis-package codes are deliberately omitted from visible headings and cells because they are repository workflow labels, not scientific study names.

Table S11 Panel A. Result-file integrity accounting by model-study cell

Panel A of Table S11. Independent accounting for the six completed model-study cells. Every cell has 600 completed arm-runs and zero unresolved duplicate, missing, unexpected or invalid records. Raw files may contain more lines than completed observations because a historical run_error record is retained when that identity is later successfully retried. Such records are provenance, not failed task outcomes, and never enter success denominators. This table therefore must not be read as a model-performance table.

Model-study cell	Expected arm-runs	Completed arm-runs	Raw result records	Historical run_error records	Duplicates	Missing	Unexpected	Invalid	Integrity status
Study 1A · Gemini	600	600	600	0	0	0	0	0	Complete; no unresolved integrity defects
Study 1A · DeepSeek	600	600	600	0	0	0	0	0	Complete; no unresolved integrity defects
Study 1B · Gemini	600	600	600	0	0	0	0	0	Complete; no unresolved integrity defects
Study 1B · Luna	600	600	602	2	0	0	0	0	Complete; no unresolved integrity defects
Study 2 · Gemini	600	600	601	1	0	0	0	0	Complete; no unresolved integrity defects
Study 2 · Luna	600	600	600	0	0	0	0	0	Complete; no unresolved integrity defects

Notes. Raw result records equal completed arm-runs plus retained historical run_error records in every cell. Study 1B Luna retains two historical run_error records and Study 2 Gemini retains one; all other cells retain none. A completed arm-run can fail its task tests and still be a valid completed observation. run_error instead means no scientific outcome was obtained for that attempt.

Table S11 Panel B. Program-level operational closure and deferred-study status

Panel B of Table S11. Program-level operational status is kept separate from scientific outcomes. Study 1A is complete by the independent recount of its two cells but predates the dedicated follow-up closure workflow. Studies 1B and 2 have explicit closure artifacts, completed all 1,200 expected arm-runs, resolved all run_error identities and stayed within their recorded budgets. Study 3 was deferred before outcome collection: NOT RUN means zero result records and no estimand, not a null result, model failure or scientific failure.

Study	Program status	Completed/expected arm-runs	Historical run_error records	Unresolved run_error identities	Recovery events	Committed/ceiling budget	Dedicated closure artifact	Scientific interpretation
Study 1A	Complete by independent cell recount	1200/1200	0	NA — no dedicated closure artifact	NA — not recorded by a closure workflow	NA — not recorded by a closure workflow	No — study predates the follow-up closure workflow	Operational recount only; not a success/failure claim
Study 1B	COMPLETE_WITH_HISTORICAL_RETRIED_RUN_ERRORS	1200/1200	2	0	NA — not separately recorded	\$1.0817566 / \$3.00	Yes	Operational closure; neither scientific nor model failure
Study 2	COMPLETE_WITH_HISTORICAL_RETRIED_RUN_ERROR	1200/1200	1	0	4	\$2.6639972 / \$5.50	Yes	Operational closure; neither scientific nor model failure
Study 3	S3-NOT-RUN (deferred to separate follow-up)	NA — no arm-runs were authorized	NA — no result collection	NA — no result collection	NA — no result collection	NA — no result collection	Terminal decision record; no result artifact	No data, no estimand, and neither scientific nor model failure

Notes. Historical run_error counts describe transport or execution attempts that were later recovered; they are not task failures and do not enter scientific denominators. NA means the relevant closure field was not recorded or is not applicable. It is never silently replaced by zero. The Study 3 decision identifier is S3-DEFER-1. No model, evaluator, Docker or benchmark call was made for that study.

Table S12. Plan formats for one task

The NL handoff plan and the JSON handoff plan produced for the same task (humanevalplus_035, Study 1A Gemini, repeat 0). In both arms the coder receives the task statement shown in the first column together with the plan. The task was chosen by a fixed rule (shortest NL plan in that cell and repeat), not for its content.

Task statement given to planner and coder	NL handoff plan (free text)	JSON handoff plan (canonical JSON)
<pre>def max_element(l: list): """Return maximum element in the list. >>> max_element([1, 2, 3]) 3 >>> max_element([5, 3, -5, 2, -3, 3, 9, 0, 123, 1, -10]) 123 """</pre>	Here is the implementation plan for the `max_element` function: **Approach:** We will use Python's built-in `max()` function, which is efficient, handles type comparisons correctly in standard use cases, and is idiomatic. **Key Steps:** 1. Directly return the result of `max(l)`. **Edge Cases to Watch For:** **Empty list:** Passing an empty list `([])` to `max()` raises a `ValueError`. We should let this default behavior occur, as standard Python collections raise an error on max of empty sets, or we can assume valid inputs based on standard HumanEval problem definitions. **Mixed or incompatible types:** Ensure the list elements are comparable (e.g., all numbers). If inputs are strictly numbers as shown in the docstring, this won't be an issue.	<pre>{ "edge_cases": ["List with negative numbers only", "List with a single element", "List with duplicate maximum values", "List containing floats and integers mixed"], "function_signature": "def max_element(l: list):", "steps": [{ "description": "Iterate through the list to find and return the maximum element.", "postconditions": ["Returns the maximum element present in the input list."], "preconditions": ["The input 'l' must be a non-empty list of comparable elements."] }], "task_id": "humanevalplus_035" }</pre>

Notes. The JSON schema requires task_id, a function_signature that starts with the task entry point, at least one step with a description and at least one postcondition, optional preconditions, and optional edge_cases. The example is illustrative and carries no measurement.

Table S13. Post-hoc schema audit of the JSON handoff arm

The contract-arm validator (agents/validator.py, unchanged) was replayed on every completed JSON handoff plan. The audit was defined after the held-out results were opened, calls no model, and is exploratory: it reports how many plans the validator would have rejected, not what a retry would have produced. Contract activations are shown for comparison.

Model-study cell	JSON handoff runs	Unparseable plans	Plans the validator would reject	Tasks affected	Plus pass among rejected	Plus pass among accepted	Contract activations
Study 1A · Gemini	150	0	0	0	—	116/150	0/150
Study 1A · DeepSeek	150	11	14	11	10/14	102/136	24/150
Study 1B · Gemini	150	0	0	0	—	114/150	0/150
Study 1B · Luna	150	0	0	0	—	110/150	0/150
Study 2 · Gemini	150	0	0	0	—	56/150	1/150
Study 2 · Luna	150	0	0	0	—	47/150	0/150

Notes. Unit: arm-run. Unparseable plans are a subset of rejected plans. Pass counts are descriptive and not a validator effect; no interval or test is computed.

Figure S1. Task-level arm outcomes in Study 1

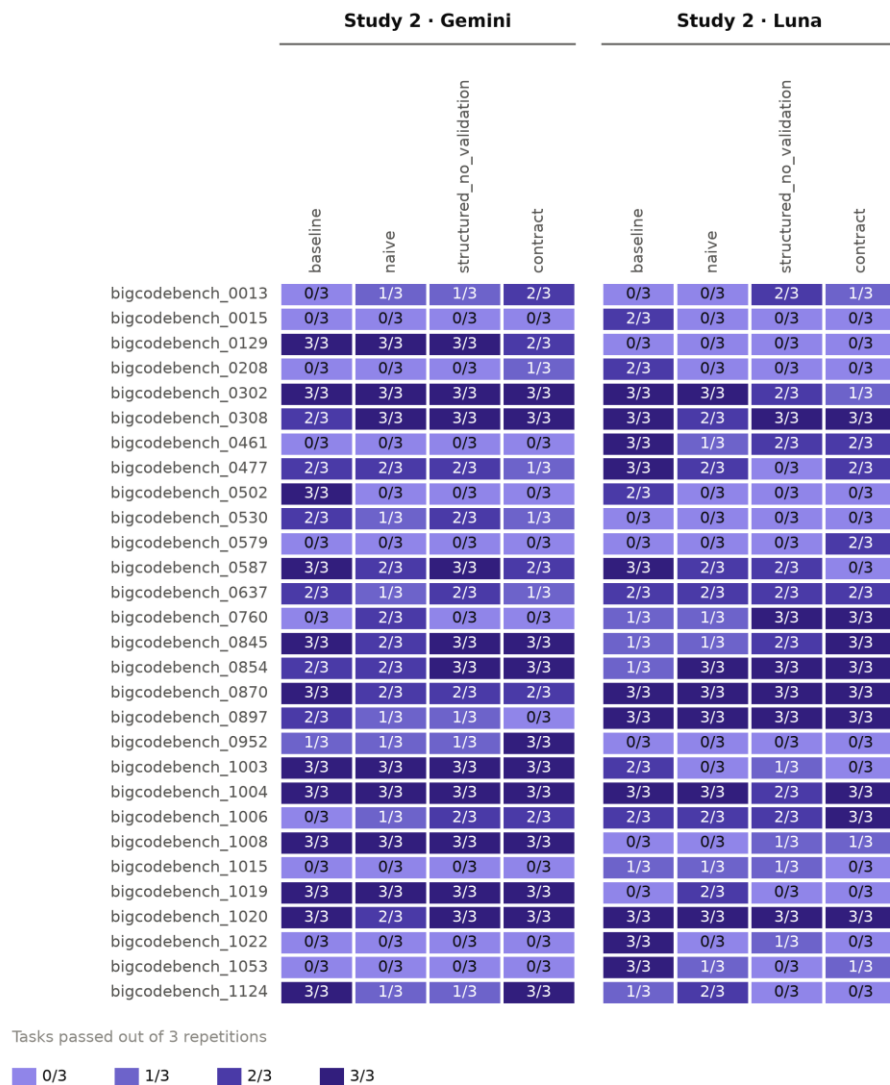
	Study 1A Gemini				Study 1A DeepSeek				Study 1B Gemini				Study 1B Luna			
	baseline	naive	structured_no_validation	contract	baseline	naive	structured_no_validation	contract	baseline	naive	structured_no_validation	contract	baseline	naive	structured_no_validation	contract
humanevalplus_008	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3	3/3
humanevalplus_011	3/3	3/3	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
humanevalplus_019	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
humanevalplus_048	3/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
humanevalplus_050	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	2/3
humanevalplus_095	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
humanevalplus_097	3/3	3/3	3/3	3/3	2/3	2/3	3/3	2/3	1/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3
humanevalplus_103	3/3	3/3	3/3	3/3	2/3	0/3	2/3	1/3	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3
humanevalplus_104	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	1/3	0/3	3/3	2/3
humanevalplus_132	0/3	2/3	3/3	2/3	0/3	0/3	0/3	0/3	1/3	1/3	1/3	1/3	0/3	0/3	0/3	0/3
humanevalplus_151	0/3	0/3	0/3	0/3	1/3	0/3	1/3	2/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	1/3
mbppplus_058	2/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
mbppplus_099	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	1/3	0/3	0/3	3/3	0/3	0/3	1/3
mbppplus_103	0/3	0/3	0/3	0/3	3/3	0/3	1/3	1/3	3/3	0/3	0/3	0/3	1/3	0/3	0/3	1/3
mbppplus_160	0/3	0/3	0/3	0/3	0/3	1/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3
mbppplus_266	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	2/3	3/3
mbppplus_308	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	1/3	3/3	3/3	3/3	3/3	2/3	2/3
mbppplus_398	0/3	0/3	0/3	0/3	0/3	0/3	0/3	1/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3
mbppplus_445	3/3	1/3	3/3	3/3	3/3	0/3	0/3	1/3	3/3	2/3	3/3	3/3	3/3	0/3	0/3	0/3
mbppplus_455	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3	2/3	3/3	3/3	1/3	2/3	2/3
mbppplus_462	0/3	0/3	0/3	0/3	1/3	3/3	1/3	1/3	0/3	0/3	0/3	0/3	3/3	3/3	3/3	3/3
mbppplus_622	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	0/3	1/3	0/3	0/3	0/3	0/3
mbppplus_745	0/3	3/3	2/3	3/3	3/3	3/3	3/3	3/3	0/3	3/3	3/3	3/3	3/3	3/3	3/3	3/3
mbppplus_796	3/3	3/3	3/3	1/3	3/3	2/3	2/3	1/3	3/3	3/3	3/3	2/3	3/3	2/3	2/3	2/3

Tasks passed out of 3 repetitions

0/3 1/3 2/3 3/3

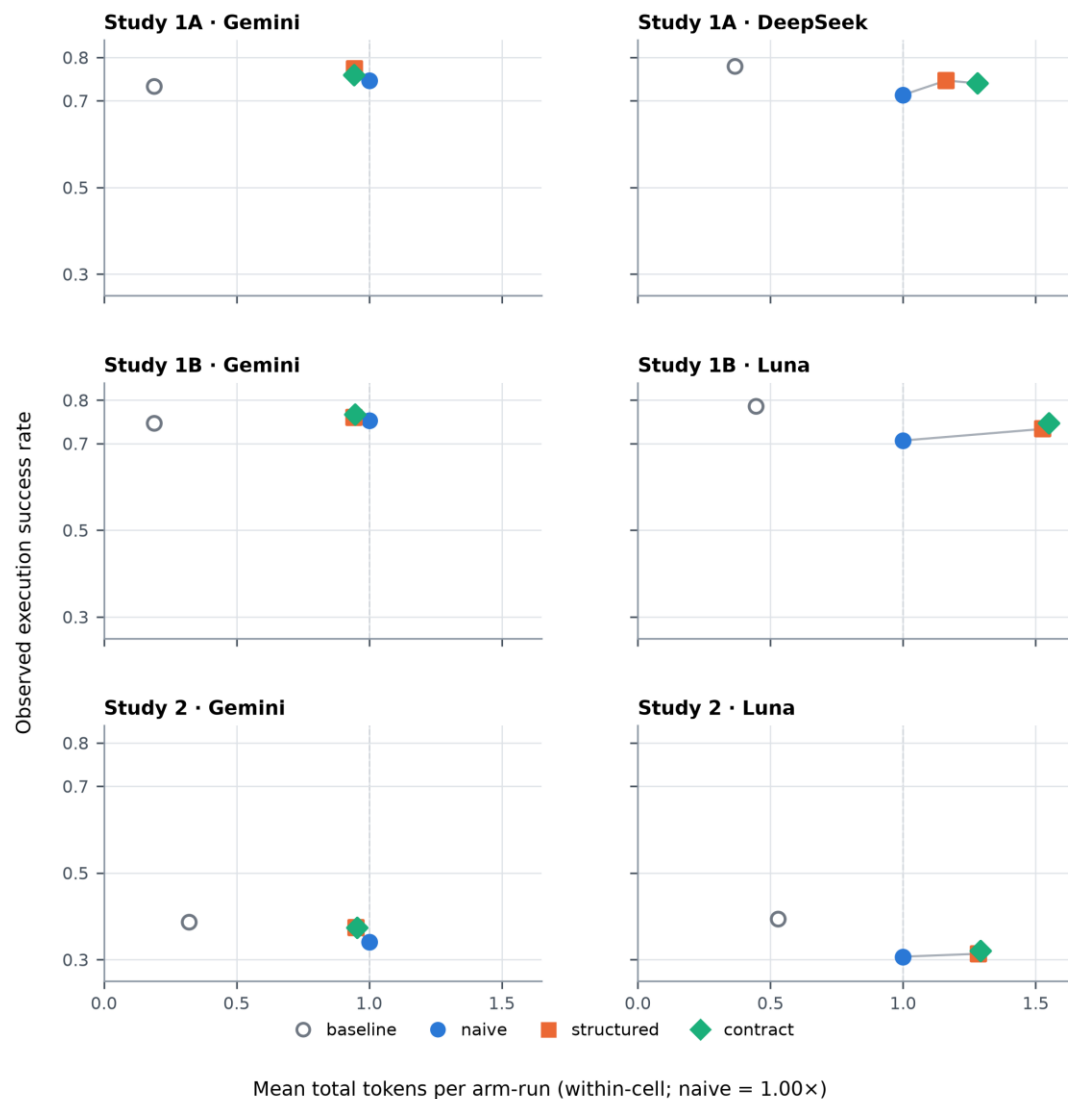
Task-level arm outcomes in Study 1. Four aligned panels, one per model-study cell, each showing the four arms in the frozen order baseline, naive, structured_no_validation, contract. Every square carries that arm's pass count out of 3 repetitions as text, so colour only speeds up reading. All four cells were run on the same EvalPlus-derived held-out task sample, so a row is the same task in every panel. The rows are the 24 of 50 tasks whose four arms do not all agree in at least one cell of this panel group. That is a deterministic post-hoc display rule, applied after the results were known and reproducible from the source table; it is not a preplanned selection and carries no inferential status. The remaining 26 tasks are omitted because all four arms returned the same pass count in every cell shown here. Where a shown task happens to agree in one panel its values are still printed, since that agreement is part of the comparison. Rows are in stable task-identifier order and are never ordered by outcome. Intermediate counts of 1/3 and 2/3 are shown as observed; the differences visible here are descriptive and are not a test.

Figure S2. Task-level arm outcomes in Study 2



Task-level arm outcomes in Study 2. Two aligned panels, one per model-study cell, each showing the four arms in the frozen order baseline, naive, structured_no_validation, contract. Every square carries that arm's pass count out of 3 repetitions as text, so colour only speeds up reading. Both cells were run on the same BigCodeBench-Hard task sample, so a row is the same task in both panels. These task identifiers are disjoint from the Study 1 sample; rows are not matched to Figure S1 and no common sample is implied. The rows are the 29 of 50 tasks whose four arms do not all agree in at least one cell of this panel group. That is a deterministic post-hoc display rule, applied after the results were known and reproducible from the source table; it is not a preplanned selection and carries no inferential status. The remaining 21 tasks are omitted because all four arms returned the same pass count in every cell shown here. Where a shown task happens to agree in one panel its values are still printed, since that agreement is part of the comparison. Rows are in stable task-identifier order and are never ordered by outcome. Intermediate counts of 1/3 and 2/3 are shown as observed; the differences visible here are descriptive and are not a test. In Study 2 the base and Plus evaluations resolve to the same result, so a cell here is not a separate hidden-test robustness measurement.

Figure S3. Within-cell execution success and token load

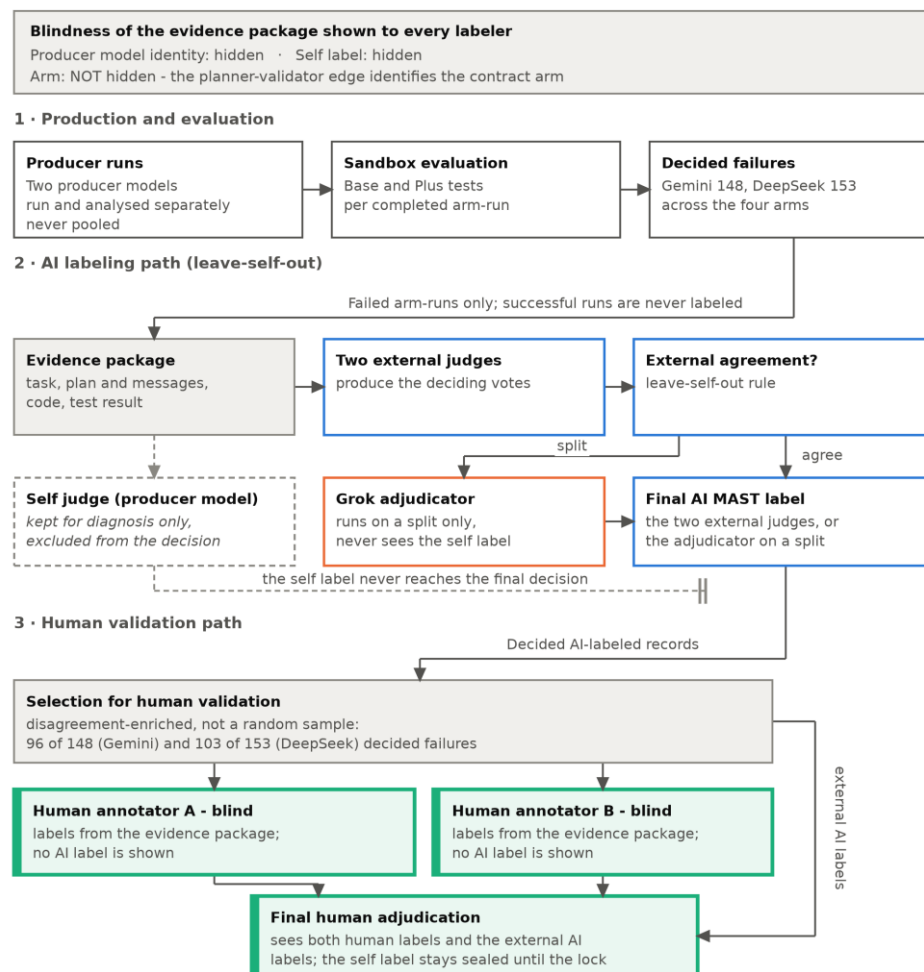


Descriptive within-cell view of observed execution success and resource load for all six model-study cells. The horizontal axis is mean input plus output tokens per completed arm-run, divided by the same cell's full-precision naive mean; raw token counts are not compared across models. The vertical axis is observed execution success rate. Only naive, structured_no_validation, and contract are connected to show the mechanism sequence. Baseline is a hollow, unconnected descriptive system reference because it is not an equal-compute control. Lines are visual guides only: the figure does not define a Pareto frontier, efficiency score, causal effect, or cross-model ranking.

Figure S4. MAST labeling and human validation workflow

MAST labeling and human validation workflow

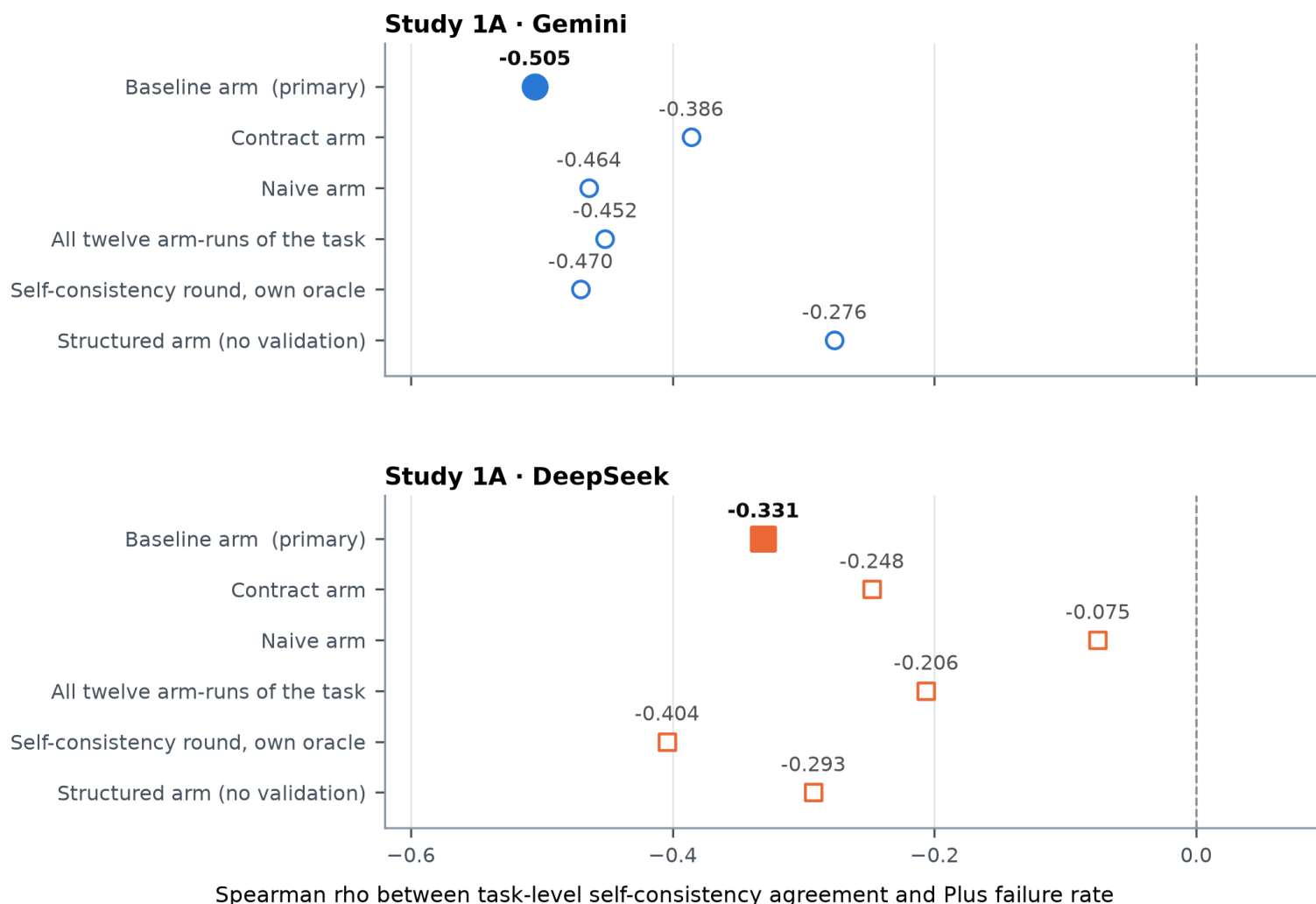
Study 1A - who produces a label, which label decides, and what each labeler can see



Human validation checks the labeling pipeline on a selected sample; it is not the source of every MAST record. The two producer models are analysed separately and are never pooled.

Labeling and validation workflow for the Study 1A MAST records. Only failed arm-runs are labeled; successful runs are never sent to a judge. The producer model's own judgement is retained as a diagnostic record and is excluded from the decision, which is why its branch ends in a barrier rather than an arrow: the final AI label is fixed by the two external judges, and a disagreement between them is resolved by the external adjudicator, which never receives the self label. Labeling is blind to producer model identity and to the self label. It is NOT blind to the arm: the planner-validator exchange appears in the contract arm alone and the plan format separates the remaining arms, so a transcript identifies its arm deterministically and arm blindness is not claimed anywhere in this work. The human round validates the pipeline on a selected, disagreement-enriched sample - 96 of the 148 decided failures for Gemini and 103 of 153 for DeepSeek - and is therefore not the source of every MAST record; agreement measured on that sample does not generalize to the whole error population. The two producer models are separate experiments and are never merged into one pipeline. Colour marks the role, never the model: every box also states its role in words, the diagnostic branch is dashed and the human path carries a thicker outline and a left accent bar, so the diagram survives greyscale reproduction.

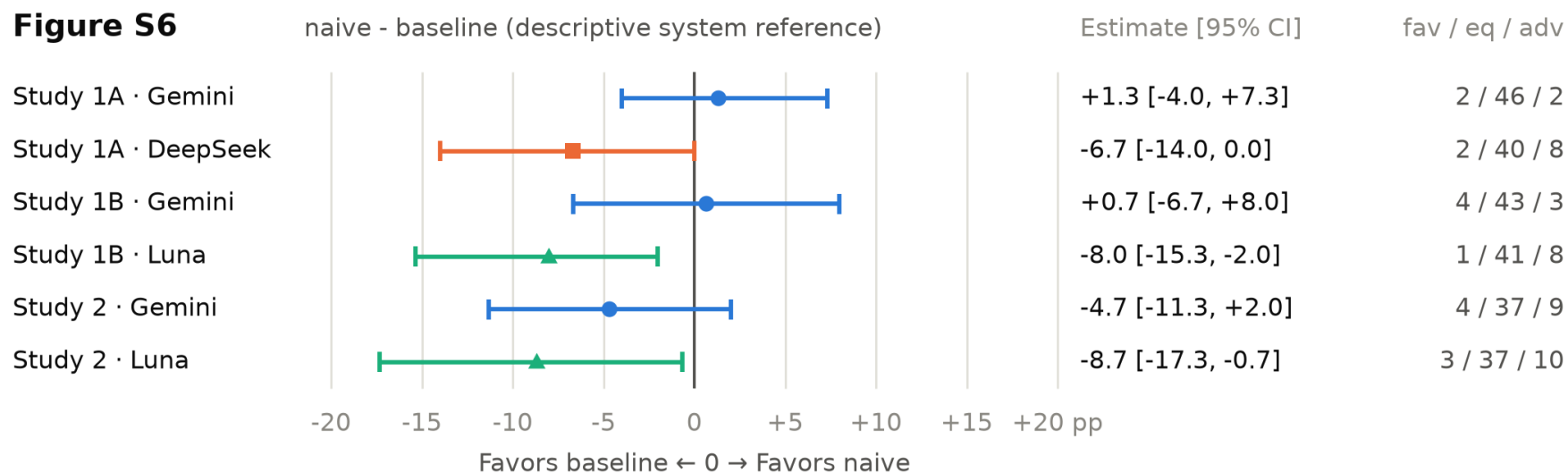
Figure S5. Task-level self-consistency and Plus failure in Study 1A



Exploratory rank correlation, 50 tasks per point; no inferential mark is attached and the two models are never combined.

Exploratory rank association between task-level self-consistency agreement and observed Plus failure, for the two Study 1A producer models. Gemini and DeepSeek occupy separate panels and share no summary value; the coefficients are never averaged or pooled. Each point is one pairing over the same 50 held-out tasks, and the baseline arm is the primary pairing because the self-consistency candidates are generated from the baseline prompt with no planner or contract layer; the other five pairings are secondary and descriptive. This display is exploratory in the sense fixed before the data were seen: no significance test, p-value or confidence interval is computed or attached, no threshold is applied, and the sign of a coefficient is not a prediction of task outcome. A negative value states only that, within this sample, tasks with more agreement among the sampled candidates also tended to fail less often; it does not license a causal reading, a screening rule or a claim that agreement forecasts correctness. The result is confined to Study 1A and does not extend to Study 1B or Study 2, which were run on different regimes and models. Agreement is a provider-independent proxy: candidates are drawn at temperature 0.8, while the arms run at the frozen default, so this is not a direct measurement of the baseline arm's own sampling distribution.

Figure S6. Descriptive system-reference contrast: naive minus baseline



Descriptive system-reference contrast for the six model-study cells: the task-level difference naive minus baseline in percentage points, with 95% percentile intervals from 10000 task-cluster bootstrap resamples. This is not one of the three mechanism contrasts and is deliberately kept out of the main forest plot. The baseline arm issues a single model call per task while naive issues two or more, so the two arms do not share a compute budget and this difference is not an equal-compute control: it cannot be read as a causal mechanism effect or as a penalty for decomposition, because the coordination mechanism and the added call budget move together. The source records this as `descriptive_only=true`, `equal_compute_control=false` and `causal_interpretation=false`. It is reported as a descriptive system reference and nothing more. Marks, typography and the favourable/equal/adverse counts follow the main forest plot; colour carries model identity only. The counts are the descriptive companion of the interval, not a second test, and cells are not pooled.